

Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies

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Abstract. VOXES is a Von Hamos X-ray spectrometer developed at the INFN National Laboratories of Frascati for high-resolution laboratory X-ray spectroscopy in the 5–20 keV range. It uses curved mosaic crystals and motorized positioning stages to perform wavelength-dispersive X-ray fluorescence (WD-XRF) with sub-10 eV tunable resolution for extended and dilute samples. Recent developments include the integration of an energy-dispersive X-ray fluorescence (ED-XRF) line based on a silicon pin-diode detector, which enables flux monitoring and simultaneous ED and WD measurements. In addition, a dedicated liquid-sample holder has been introduced, and a Y-shaped support geometry, crucial for switching to a transmission layout, provides mechanical compatibility with laboratory XAS, now under implementation. These upgrades expand the versatility and automation of VOXES, strengthening its role as a table-top platform for laboratory X-ray spectroscopy.

Introduction

X-ray spectroscopy [1], in its various forms, offers a well-established method for probing the state of matter under different physical and chemical conditions. Different X-ray techniques offer specific capabilities and applications across diverse fields such as materials science [2], chemistry and catalysis [3] with applications for battery research [4, 5], gemmology [6], cultural heritage [7], environmental analysis [8], and industrial quality control [9].

Among the techniques that detect X-rays, core-hole spectroscopies [10], such as X-ray Absorption Spectroscopy (XAS) and X-ray Emission Spectroscopy (XES), play a central role in probing the local electronic environment of atoms. The creation of a deep-core vacancy and its subsequent relaxation encode detailed information about oxidation state, spin configuration, ligand environment, and bonding characteristics [11].

Traditionally, these measurements mostly have been performed at large-scale facilities, such as synchrotrons and X-ray free-electron lasers, which provide highly brilliant and tunable X-ray beams [12]. Despite their remarkable performance, access to these infrastructures remains limited due to high demand, restricted beamtime availability, and the logistical constraints associated with conducting experiments in a facility. This has driven a growing interest in developing laboratory-scale X-ray spectroscopy setups [13, 14] capable of delivering high-resolution measurements using compact and cost-effective X-ray sources like X-ray tubes [15]. For instance, several table-top setups have been demonstrated for XAS and XES measurements [16], including configurations capable of performing both simultaneously [17, 18].

In this scenario, the VOXES [19, 20] spectrometer at the INFN National Laboratories of Frascati represents an example of how optimized optical design, advanced detectors, and precise motorized positioning systems can significantly enhance the performance of laboratory XES for large sources. Originally developed for application of kaonic atom X-ray spectroscopy at DAΦNE [21] within the SIDDHARTA-2 experiment [22], VOXES has evolved into a versatile platform capable of measuring emission spectra from extended or dilute samples, with a tunable sub-10 eV energy resolution in the 5–20 keV range [23].

In this article, we present the recent developments of the VOXES spectrometer. We first describe the main features of the setup and its operation in emission geometry for extended sources. We then introduce the design upgrades implemented to improve sensitivity and versatility. Finally, we show

how the system can be reconfigured between emission mode (XES) and transmission mode (XAS), enabling both types of measurements within the same compact platform. The paper is structured as follows. Section 2 describes the operating principle of the VOXES setup. Section 3 presents the updated configuration, which allows switching between XES and XAS modes. Section 4 concludes the paper.

VOXES Experimental Setup

The VOXES experimental setup is shown in Fig. 1. Compared to previous publications [23], the setup now consists of two complementary detection lines: a Wavelength-Dispersive (WD)-XRF line based on a Von Hamos geometry with a mosaic crystal, and an Energy-Dispersive (ED)-XRF line using a compact silicon detector. The entire spectrometer is mounted on a motorized rail system to ensure reproducible positioning and remote operation.

The X-ray source is a MicroXray WindChill tube with a molybdenum anode. The Mo $K\alpha$ line at

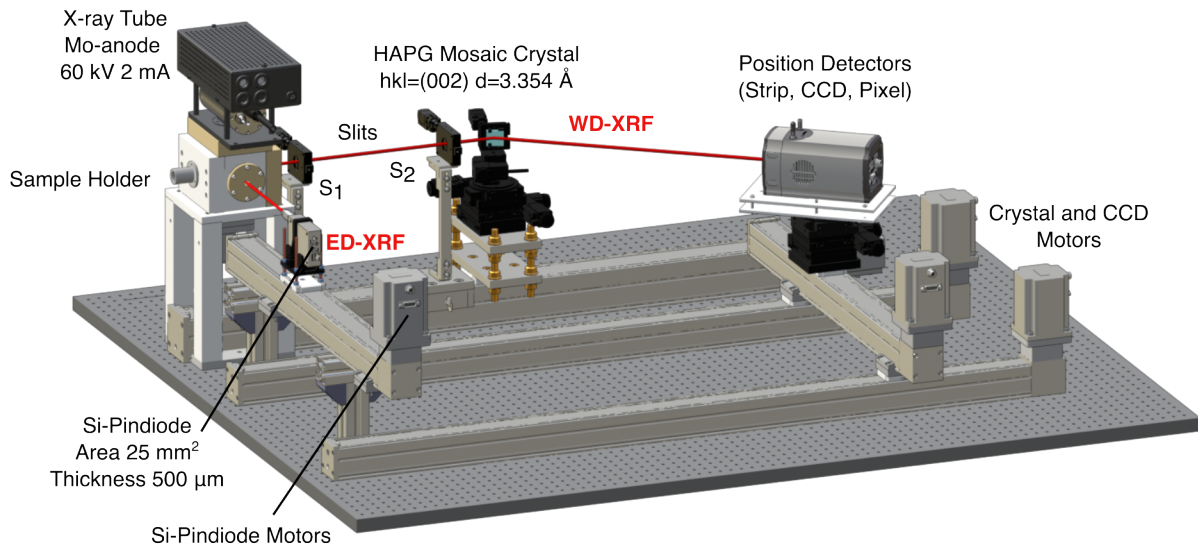


Fig. 1: The VOXES experimental setup. The X-ray tube on the left directs the beam onto the sample holder. After emission, the beam is conditioned by adjustable slits and motorized stages to ensure precise alignment. The resulting X-rays are diffracted by the mosaic crystal and subsequently detected by the position detector placed on the right.

17 keV results in a peak-free continuum in the 5–12 keV range, that is ideal for studying $3d$ transition metals without source-line contamination. The WD-XRF line consists of a slit system, a curved mosaic crystal, and a position-sensitive detector. Two motorized STANDA slits (S_1 and S_2) shape the beam emerging from the source box and define the effective source size and angular acceptance. Typical settings are an effective source width of $S'_0 = 1$ mm and an angular acceptance $\Delta\theta' = 0.4^\circ$ [20], allowing a controlled trade-off between spectral resolution and photon flux. The mosaic crystals used in VOXES are highly annealed pyrolytic graphite (HAPG) crystals, all oriented along the (002) direction with a lattice spacing of $d = 3.354$ Å. They share the typical properties of HAPG, namely high reflectivity, low absorption in the 5–20 keV range, and a controlled mosaic spread that enables efficient diffraction of divergent beams, with a declared mosaicity of $0.10^\circ \pm 0.01^\circ$. The specific crystals available for VOXES differ in their bending radius and thickness, allowing the instrument to be optimized for different applications. A set of cylindrical crystals is available with radii $\rho_c = 10.3$ mm,

77.5 mm, 103.4 mm, and 206.7 mm, as well as thicknesses of 20 μm , 40 μm , and 100 μm . The crystal mount is actuated by a five-axis STANDA positioning system, enabling fine control of its position and orientation with respect to the Bragg angle θ_B .

The diffracted spectrum is recorded with a position-sensitive detector. Several detectors are available depending on the target energy range and efficiency requirements. For one-dimensional detection, the setup can employ a MYTHEN2 strip detector (DECTRIS) with a 50 μm strip pitch and an active area of $32 \times 8 \text{ mm}^2$. When two-dimensional imaging is required, particularly for applications where the vertical alignment of the dispersed spectrum is critical, an ANDOR iKon-M CCD camera can be used, featuring a 1024×1024 pixel sensor with a $13 \times 13 \mu\text{m}$ pixel pitch and an active area of approximately $13.3 \times 13.3 \text{ mm}^2$. Alternatively, a Timepix detector with a 2×2 module configuration is available, providing a total active area of 512×512 pixels with a 55 μm pixel size. All position-sensitive detectors are mounted on a motorized linear stage combined with a fine rotation axis, enabling precise positioning and optimal focusing of the Bragg-diffracted spectrum along the dispersion direction.

FESTO electrical stage motor and STANDA stepper motors provide full motorization of the slit sys-

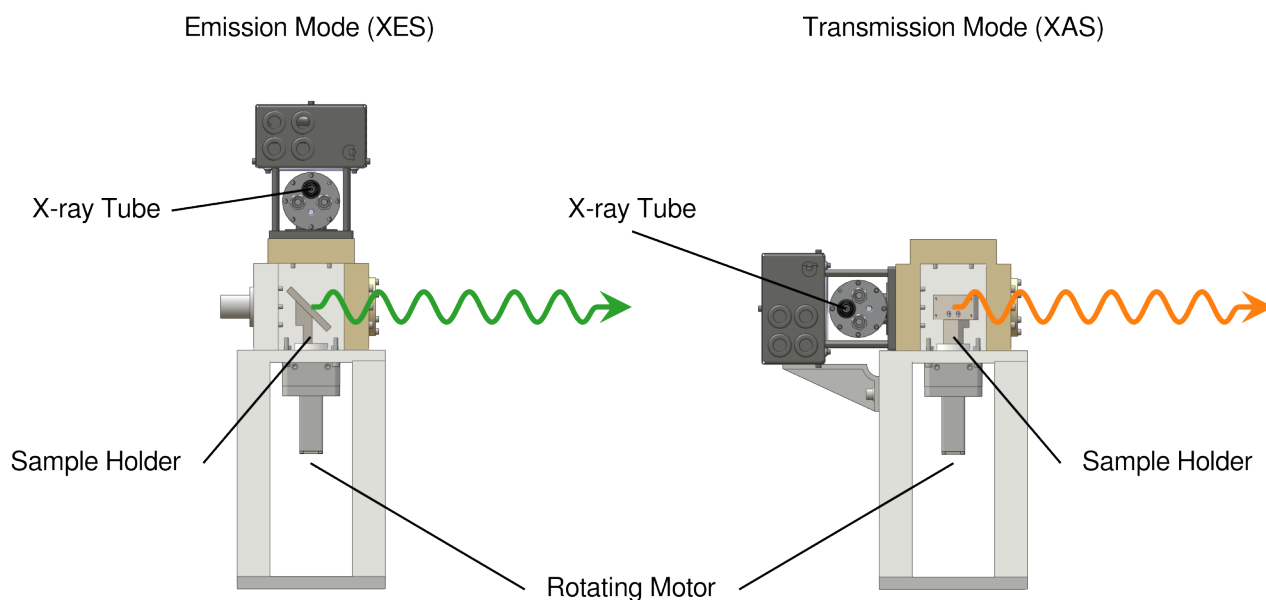


Fig. 2: VOXES spectrometer in transmission (left) and emission (right) modes. In transmission mode, the beam passes through the sample for XAS measurements. In emission mode, the emitted X-rays are detected for XES analysis. The setup allows switching between the two modes by rotating the source and detector assemblies.

tem, crystal mount, and detector stage. This ensures stable, repeatable alignment during scans and allows remote operation during long measurements. Fine adjustments of the Bragg angle and detector position are controlled through the motorized five-axis stages, which also provide pitch and roll corrections to maintain optimal focusing.

The ED-XRF line can operate in parallel with the WD-XRF system and employs an Amptek X-123 silicon pin-diode with integrated preamplifier and digital pulse processing. The detector uses a 25 mm^2 active area with approximately 500 μm silicon thickness and a thin Be window ($\approx 12.5 \mu\text{m}$) for optimal transmission of low-energy X-rays. It delivers an energy resolution of 125–140 eV FWHM at 5.9 keV. The sample holder is mounted on a motorized rotation stage, allowing the sample to be oriented toward either the WD-XRF or ED-XRF line. It includes a Y-shaped aperture, which is used for optical alignment in emission geometry and also enables transmission of the X-ray beam when

Table 1: Position-sensitive detectors available in the VOXES spectrometer and their main characteristics and intended use cases.

Detector	Type	Advantages	Limitations	Applications
MYTHEN2	1D Strip	High count-rate and fast readout	No vertical information	Routine WD-XRF, calibration, high-statistics studies
ANDOR iKon-M	2D CCD	High spatial resolution, low noise	Limited count rate, slow readout	High-resolution XES, alignment and imaging
Timepix	2D Pixel	Single-photon sensitivity, flexible 2D imaging	Lower count rate	Low-flux measurements

the setup is reconfigured for XAS for future developments. The holder accepts custom mounts for different sample types; dedicated Teflon cells with thin Kapton windows were fabricated for liquid samples, enabling measurements of aqueous or organic solutions without realigning the spectrometer.

VOXES Operating Modes

The VOXES spectrometer can operate in multiple configurations depending on the measurement requirements. Its fully motorized geometry allows switching between WD-XRF and ED-XRF, with minimal manual intervention like shown in Fig. 2. The modular design also enables a transmission configuration that provides compatibility with XAS measurement, currently under development. In emission mode, the sample is positioned on the sample-holder at 45° so that it is directly irradiated by the X-ray tube. The emitted X-rays then impinge on the mosaic crystal, where their distance depends on the specific Bragg angle for the emission line under study. The mosaic crystal diffracts the photons, and the dispersed spectrum is recorded by the position-sensitive detector, one-dimensional with a strip detector or two-dimensional with a CCD or pixel detector. Motorized slits (S_1 and S_2) allow adjustment of the effective source size and angular acceptance, allowing a balance of energy resolution versus counting rate, depending on the target requirements. This geometry is used for high-resolution studies of emission lines and for laboratory WD-XRF measurements.

An example of the VOXES WD-XRF operating modes is shown in Fig. 3 obtained with the MYTHEN2 detector. The left panel displays a spectrum acquired from a FeCoNi metallic foil. This foil is used during alignment and energy calibration, taking advantage of the well-separated Co $K_{\alpha 1,2}$ and Fe K_{β} lines. Once the setup is aligned and calibrated, the same geometry can be used to investigate the electronic configuration of iron-containing samples. The right panel shows two preliminary XES measurements on iron compounds in different spin states: high-spin for FeSO_4 and low-spin for K_4FeCN_6 . Spectra were acquired using the X-ray tube at 25 kV and 2 mA, with a total acquisition time of 4 hours. Due to the different spin configuration of iron, high and low respectively, the $K_{\beta'}$ satellite is strongly suppressed in the low-spin compound, while it is clearly visible in the high-spin case [24] and a few eV shift. This demonstrates the ability of the VOXES spectrometer to resolve chemical and spin-state dependent features in the Fe $K_{\beta 1,3}$ emission line.

The ED-XRF line can operate independently and in parallel with the WD-XRF system. The silicon pin-diode detector is mounted on a motorized stage that allows it to be positioned closer to the sample to increase the detected signal, or retracted to avoid excessive dead time when the emitted flux is high. It collects energy-dispersive fluorescence spectra in the 2–25 keV range. This mode is typically used to monitor the emitted flux, particularly from dilute samples such as liquids, or as a complementary detection channel for quantitative analysis. The sample holder includes mounts for both solid and liquid samples (see Fig. 4). Liquid measurements are carried out using dedicated Teflon pouches with external dimensions of $25 \times 80 \times 2.5 \text{ mm}^3$ and an exposed active area of $15 \times 60 \text{ mm}^2$, encapsulated with $7 \mu\text{m}$ thin Kapton windows. This thickness results in a transmission of approximately 40% at 7

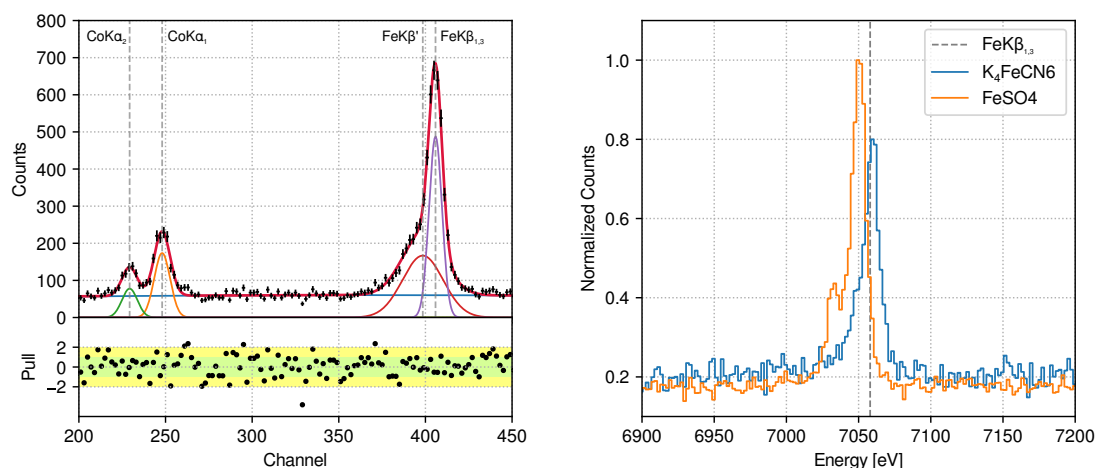


Fig. 3: WD-XRF Spectrum of a FeCoNi foil (left) used to align and calibrate with cobalt and iron reference line. Spectra (right) of high-spin K_4FeCN_6 (blue) and low-spin $FeSO_4$ (orange). The Fe K β' suppression in low-spin compounds is clearly resolved, indicating sufficient sensitivity for chemical-state analysis.

keV for water-like liquid solutions, which is close to the optimal regime for transmission-mode XAS measurements [13]. Other path lengths are available and can be selected depending on the photon energy and sample composition. The cells can be installed without modifying the optical alignment, allowing the system to switch rapidly between solid and liquid samples while preserving the same measurement geometry. The VOXES setup has been mechanically upgraded to allow operation in a transmission geometry suitable for laboratory XAS. This configuration is enabled by a Y-shaped sample holder and a reversible X-ray tube mount, which allows the beam to illuminate the sample from behind. Although XAS measurements are still under development, the hardware required to perform absorption studies is already in place and will be validated in future work.

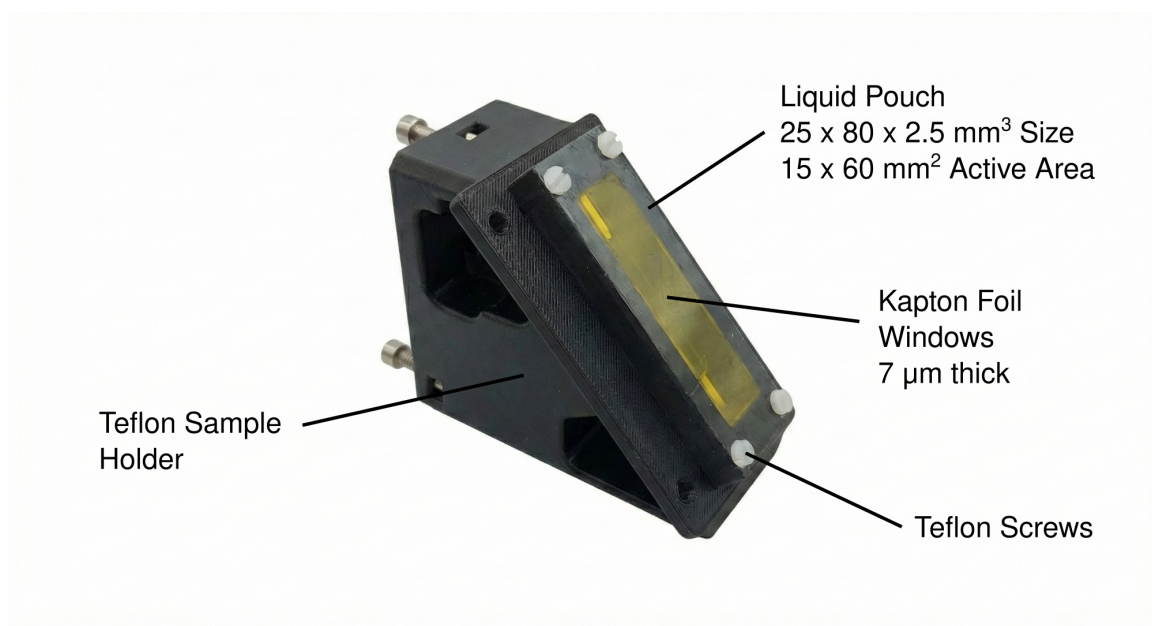


Fig. 4: Teflon liquid-sample holder used for solution measurements. The liquid is contained in a flexible pouch fixed with Teflon screws to prevent metal contamination, and sealed with a Kapton window to ensure X-ray transmission.

Conclusion and Outlook

In this work, we presented the recent developments of the VOXES spectrometer at the INFN National Laboratories of Frascati. The setup now features full motorization of all key components, including slits, crystal and detector, ensuring stable, reproducible, and remotely controlled operation for laboratory X-ray spectroscopy. The implementation of a dedicated sample holder, with the possibility to perform measurements on liquids, expands the range of measurable materials to include solutions and non-solid samples. In addition, the integration of a complementary ED-XRF line based on a silicon pin-diode enables fast alignment, flux monitoring, and simultaneous WD+ED acquisition, significantly improving workflow efficiency. The mechanical layout has also been upgraded to allow operation in a transmission mode geometry. Through the introduction of a Y-shaped sample holder and a reversible tube mount, the system is now compatible with transmission mode measurements for acquiring XAS spectra. These improvements increase the flexibility and usability of the VOXES spectrometer, reinforcing its role as a compact and versatile platform for laboratory-scale XRF measurements. Future work will focus on completing the XAS implementation and further optimizing the automation and data-acquisition framework.

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